Cosmology

Leonard Susskind

Core track, 2009 Winter Legacy Cosmology
Transcript-derived notes curated by LazyingArt LLC with Video2Book

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC with Video2Book.

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Chapter 1

Geometry of the Expanding Universe

These notes follow Leonard Susskind's first cosmology lecture as closely as possible in both argument and pace. The transcript, together with selected blackboard frames preserved through LazyingArt LLC and Video2Book, lets us keep the order of discovery clear: first a toy expanding universe, then Hubble's law, then the metric of space, and only then the metric of spacetime. The lecture is deliberately geometry-first. Newtonian cosmology, Einsteinian dynamics, and the full Friedmann–Robertson–Walker package are promised for later. Here we are interested in a simpler question: what does it mean, geometrically, to say that the universe expands?

1.1 Geometry First, Dynamics Later

Susskind begins by declining to tell the whole history of cosmology. We are to get straight to the point, and the point is the expanding universe and its geometry. That opening is not merely rhetorical. It tells us why this chapter begins with idealized lines, grids, and metrics, and not with Einstein's equations.

The roadmap is brief but important. We will study the geometry of expanding universes first and leave the physics of expansion for later. Newtonian $F = ma$ cosmology will come next, and only after that will geometry and dynamics be fused into the full FRW description. The present lecture therefore asks us to be disciplined: for now we describe the universe; we do not yet ask what drives the description.

1.2 Comoving Coordinates and the One-Dimensional Expanding Line

We begin with the simplest possible space: one-dimensional space, *space, not spacetime.* Take an infinite line, place equally spaced galaxies on it, and label the galaxies by a coordinate x. The first conceptual point is that the coordinate is not yet a measured distance. It is a tag attached to a galaxy, and it moves with the galaxy if the universe expands or contracts. One may think of x as the name of the galaxy.

Suppose the distance from the galaxy called $x = 0$ to the galaxy called $x = 1$ is some measured quantity a. Then the distance between any two galaxies is the coordinate difference multiplied by

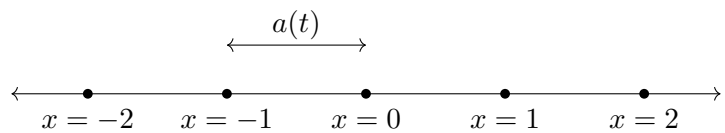


Figure 1.1: A one-dimensional universe of equally spaced galaxies. The labels $x = -2, -1, 0, 1, 2$ are comoving coordinates; the neighboring physical separation is the scale factor $a(t)$.

this common spacing:

$$d = a\,\Delta x. \tag{1.1}$$

This is the first mathematical move that matters. The coordinate difference Δx is not itself the physical distance. It only records which two galaxies we have chosen. The scale factor a converts a comoving label difference into a true geometric separation.

Now we imagine the line as an elastic thread. Two giants pull on its ends and stretch it or let it contract. The labels remain attached to the galaxies, but the neighboring physical spacing changes. Thus a becomes time-dependent:

$$d(t) = a(t)\,\Delta x. \tag{1.2}$$

At this stage homogeneity means simply that neighboring galaxies are equally spaced everywhere. The common spacing may still vary with time.

This is already enough structure to say something nontrivial. We have a universe in which points carry fixed comoving labels while the physical distance between them changes. That is the raw material from which the rest of the lecture is built.

1.3 From the Scale Factor to Hubble's Law

We now calculate the rate at which the distance between two fixed galaxies changes. The word fixed matters: we keep the same pair of galaxies, so Δx is constant in time. Differentiating $d = a\,\Delta x$ gives

$$\dot{d} = \dot{a}\,\Delta x \tag{1.3}$$

$$= \frac{\dot{a}}{a}\,a\,\Delta x \tag{1.4}$$

$$= \frac{\dot{a}}{a}\,d. \tag{1.5}$$

This is Hubble's law in its cleanest form. If we define

$$H(t) \equiv \frac{\dot{a}}{a}, \tag{1.6}$$

then the law reads

$$\dot{d} = H(t)\,d. \tag{1.7}$$

Susskind follows the standard historical language and says "Hubble constant," but he immediately pauses to correct the impression that the name gives. The quantity is constant in space, not in time. It is the same for every galaxy at a given moment, but in general it evolves with the universe. So in the notes we should hear his warning and keep in mind that $H(t)$ is really the Hubble parameter.

The physical content comes next. Galaxies farther away recede faster. But the argument does not pick out a center. Nothing in the formula privileges $x = 0$, $x = 1$, or any other point. The law concerns relative separation, and it is the same everywhere. The expanding universe described here is therefore not a blast from one distinguished place.

Only after deriving the law does the lecture turn outward toward observation. If Alice and Bob exchange light signals while moving apart, the received light is redshifted. The faster the recession, the greater the redshift. Spectral lines therefore offer a way to estimate recession speed. Distance is harder, but Hubble used rough observational surrogates such as brightness and angular size. His plot was not a perfect straight line. It was noisy, with large scatter, and his original distances were badly misestimated. But the geometric principle was right: recession speed is proportional to distance.

A second caution is briefly raised and deferred. If d is large enough, the formula predicts $\dot{d} > 1$ in units with $c = 1$. Susskind flags that as important, but leaves the resolution for later lectures.

1.3.1 Question & Answer

Question. Does $v = Hd$ mean that a given pair of galaxies must be accelerating, simply because the distance d is increasing?

Answer. No. The law says that velocity increases with distance at a given moment. It does *not* by itself say whether the velocity of one fixed pair increases with time. That depends on how $a(t)$ behaves.

If

$$a(t) = \alpha t, \tag{1.8}$$

then neighboring galaxies separate at constant speed:

$$\dot{d} = \alpha \, \Delta x. \tag{1.9}$$

The recession speed depends on the pair we choose, but it does not change with time.

If instead

$$a(t) = \beta t^2, \tag{1.10}$$

then

$$\dot{d} = 2\beta t \, \Delta x, \tag{1.11}$$

so the recession speed of a fixed pair increases with time.

And if

$$a(t) = \gamma \log t, \tag{1.12}$$

then

$$\dot{d} = \frac{\gamma}{t} \, \Delta x, \tag{1.13}$$

so the recession speed decreases with time.

Thus Hubble's law is always true as a function of distance. Whether the universe is accelerating or decelerating is a second question, encoded in the detailed time dependence of $a(t)$. Susskind uses exactly these elementary examples to separate the two issues before moving on.

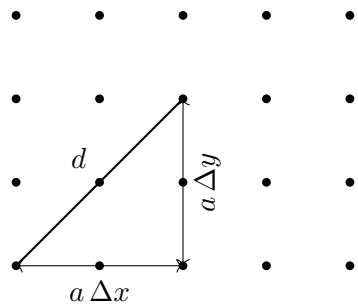


Figure 1.2: A two-dimensional comoving grid. The lecture uses such a grid only as a geometric idealization, but it makes the roles of homogeneity, isotropy, and the Pythagorean theorem transparent.

1.4 Higher Dimensions, Isotropy, and Homogeneity

At this point the lecturer says, in effect, that there is nothing especially one-dimensional about the argument. So let us widen the stage. In two dimensions we imagine a square grid of galaxies. Each galaxy carries two labels, (x, y), and these labels again move with the galaxies rather than measuring physical distance directly.

Before writing the distance formula, the lecture asks a natural question: why should the spacing along the x-direction be the same as the spacing along the y-direction? The answer is not abstract formalism; it is observation. If the galaxy distribution were systematically more crowded along one axis than another, we would see that anisotropy in the sky. On sufficiently large scales, we do not. That empirical fact is what motivates using the same scale factor in all spatial directions.

With that understood, the distance between two galaxies separated by Δx and Δy is found from the Pythagorean theorem:

$$d = a\sqrt{(\Delta x)^2 + (\Delta y)^2}. \tag{1.14}$$

And in three dimensions the extension is immediate:

$$d = a\sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}. \tag{1.15}$$

For a fixed pair of galaxies the coordinate differences are constant, so differentiation once again gives

$$\dot{d} = \frac{\dot{a}}{a}\, d. \tag{1.16}$$

The Hubble law survives unchanged.

Now the lecture names the two symmetries that have been guiding the construction. Homogeneity means sameness from place to place. Isotropy means sameness from direction to direction. Susskind is careful not to over-polish this into textbook dogma. The universe is not homogeneous on the scale of a galaxy, nor even on modest astronomical scales. It is not perfectly isotropic when we look within the Milky Way. But after averaging over many galaxies, the large-scale sky looks roughly the same in every direction, and that is the working content of the cosmological principle at this stage.

1.5 Why the Scale Factor Is a Bookkeeping Device, and Why the Metric Matters

The lecture next doubles back to a subtle point that is easy to miss if we move too fast. If $H = \dot{a}/a$ is what enters Hubble's law, does the overall numerical value of a by itself have any physical meaning? Susskind's answer is no. The scale factor is, in part, a bookkeeping device. One may redefine it by a constant multiple without changing the observable Hubble law.

If we introduce a new scale factor b by

$$b = 3a, \tag{1.17}$$

then

$$\frac{\dot{b}}{b} = \frac{3\dot{a}}{3a} = \frac{\dot{a}}{a}. \tag{1.18}$$

So the overall normalization of the scale factor is conventional. What matters physically is how it changes, not its absolute magnitude. This is the lecture's answer to the question about the "initial value" of a: one may choose units or normalization freely, but the ratio $\dot{a}/a$ does not care.

Only after that clarification does Susskind turn from kinematics to geometry proper. For neighboring points we replace Δx by dx. Then the physical distance between neighboring points is

$$ds = a\,dx, \tag{1.19}$$

and therefore

$$ds^2 = a^2\,dx^2. \tag{1.20}$$

This is now a spatial metric. In one dimension the coefficient of dx^2 is simply

$$g_{xx} = a^2. \tag{1.21}$$

So an expanding universe can be described by saying that the spatial metric itself is time-dependent.

The lecture then pushes one step further and asks for the spacetime metric. Proper time must include both the time separation and the physical spatial separation. Since the physical spatial distance is $a\,dx$, the natural one-dimensional spacetime line element becomes

$$d\tau^2 = dt^2 - a^2(t)\,dx^2. \tag{1.22}$$

For a comoving observer, $dx = 0$, so proper time reduces to ordinary time. For an object moving across the comoving grid, the spatial term contributes in the usual relativistic way.

The frame below is worth keeping precisely because it shows how the lecture makes the transition on the board: spacetime sketch on the left, cleaned line element in the middle, metric notation on the right.

In three spatial dimensions the board writing is only partly visible, so the standard clean form should be read as a cautious reconstruction supported by both the transcript and the frame:

$$d\tau^2 = dt^2 - a^2(t)\big(dx^2 + dy^2 + dz^2\big). \tag{1.23}$$

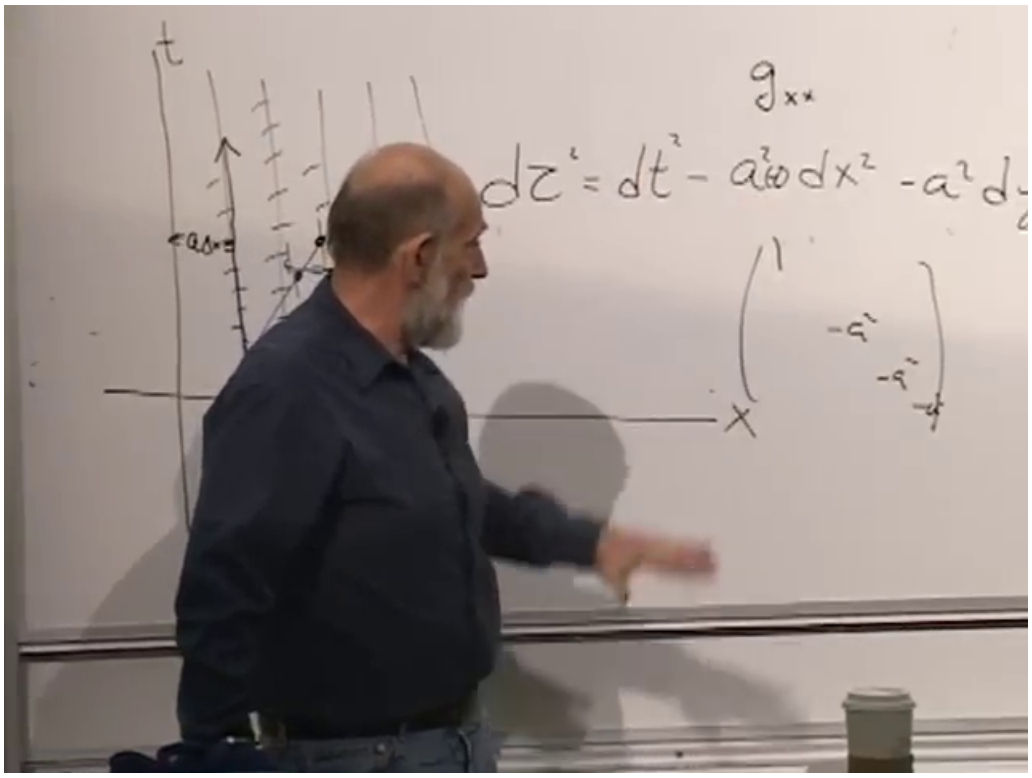
Likewise, the matrix on the right side of the board is partly occluded, but the intended diagonal form is clear enough to write it carefully as

$$g_{\mu\nu} = \mathrm{diag}\big(1, -a^2, -a^2, -a^2\big). \tag{1.24}$$

The off-diagonal terms vanish:

$$g_{tx} = g_{ty} = g_{tz} = 0, \tag{1.25}$$

which is the lecture's way of saying that in this simple ansatz there are no mixed $dt\,dx$-type terms.



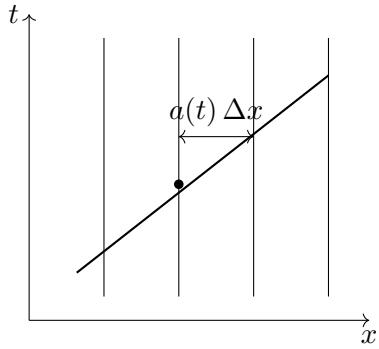


Figure 1.3: Expanding-universe spacetime sketch and metric evidence. The screenshot preserves the lecture's board layout; the redraw isolates the comoving worldlines and the fixed-time spatial separation.

1.5.1 Question & Answer

Question. Is dx a physical distance, or only a comoving label?

Answer. It is only a comoving label interval. The physical distance is $ds = a\,dx$. The clean way to think about it is exactly the way Susskind phrases it: draw a coordinate mesh through the galaxies and decree that the mesh follows them. Then Δx between two chosen galaxies stays fixed by construction, while the measured distance between them changes because the scale factor changes. One may place the units in x, or in a, or split them between the two; that is convention. What is not conventional is the product $a\,dx$, and what is especially invariant is the ratio $\dot{a}/a$.

The lecture also pauses here over another convention. The coefficient of dt^2 has been set equal to 1, and Susskind remarks that if that coefficient depended only on time, one could absorb it into a redefinition of the time variable. Similarly, he adopts units with $c = 1$. None of that changes the main point: expansion is the time dependence of the spatial components of the metric.

We are still not doing dynamics. One could now ask what energy-momentum tensor inserted into Einstein's equations would produce this metric, but the lecture explicitly postpones that. Geometry first, dynamics later.

1.6 What We Know, What We Assume, and What the Horizon Hides

With the metric in hand, Susskind makes what he calls a slightly finer distinction. Homogeneity means the metric coefficients do not depend on spatial position. Isotropy means the spatial directions are treated alike. In the diagonal metric above, the fact that the coefficients do not depend on x, y, or z is the metric statement of homogeneity. The fact that the three spatial coefficients are equal is the metric statement of isotropy.

1.6.1 Question & Answer

Question. What exactly do homogeneity and isotropy buy us, and how are they related?

Answer. They buy us an enormous simplification. Homogeneity says the metric can depend on time but not on where we are in space. Isotropy says the x-, y-, and z-directions must enter in the same way. The two ideas are not identical: a universe may be homogeneous and still have a preferred direction. But isotropy about every point forces homogeneity. That is why the lecture treats them as distinct concepts that are closely related rather than identical principles.

The lecture then becomes deliberately cautious. The universe is not exactly homogeneous at small scales, nor exactly isotropic in every local observation. A farmer's field provides the analogy: on one scale it looks homogeneous, on a smaller scale it plainly does not, and on a scale larger than the whole field it again ceases to look homogeneous. Cosmology is similar. On scales of galaxies the universe is highly structured; only after averaging over much larger scales does the rough uniformity emerge.

That caution widens into a discussion of the horizon. We do not see the whole universe. More

strongly, we do not even see all of the universe that could ever in principle affect us. Beyond the observable region we do not know whether homogeneity and isotropy continue to hold. There may be indirect clues, but the lecture insists that the horizon is an ultimate barrier to direct knowledge.

This is why the cosmological principle is both fruitful and modest. It is a powerful empirical approximation on the scales astronomy can probe, but it is not a metaphysical guarantee about arbitrarily large scales.

1.7 Closed and Bounded Universes: The Circle Model

Only after laying out the flat expanding picture does the lecture pivot to closed geometries. The order matters. We are now varying the global shape of space while preserving the local logic of comoving labels, scale factor, and Hubble flow.

The first step is to reject the wrong bounded models. A line segment has finite extent, but it comes with distinguished endpoints. A cube with walls is finite, but it violates both homogeneity and isotropy: observers near a wall see something different from observers far from a wall, and even an observer at the center sees different things in different directions. So boundedness, if the universe is to remain homogeneous and isotropic, cannot mean a box with edges.

The right idea is bounded without boundary. Susskind first gestures toward the surface of a sphere, not the interior of a ball. A two-dimensional creature confined to the surface of a featureless sphere finds every place and every direction locally the same, yet the world is finite. That is the model of boundedness he wants. He then reduces the example to one dimension, because the algebra is simpler there: life on a circle.

In one dimension isotropy means left and right are equivalent. A circle has exactly that feature. The convenient coordinate is the angle θ, measured in radians. We choose one galaxy and call it $\theta = 0$, then label every other point by an angular coordinate. A point just to the "left" of the origin may be described by a small negative angle or, equivalently, by an angle just short of 2π. That periodicity is part of the geometry and not a side remark.

There is also a brief notational wobble in the lecture that is worth preserving in cleaned form. Susskind momentarily hesitates over whether to call the whole circumference the scale factor or to write it as 2π times a radius-like quantity. The cleaner convention, and the one we keep, is

$$L = 2\pi a, \tag{1.26}$$

where L is the total distance around the circle and a is the radius-like parameter of the external plotting circle.

Then, for a small angular separation $\Delta\theta$, the physical arc length is

$$d = a\,\Delta\theta. \tag{1.27}$$

Passing to infinitesimals gives the local metric

$$ds = a\,d\theta, \tag{1.28}$$

and therefore

$$ds^2 = a^2\,d\theta^2. \tag{1.29}$$

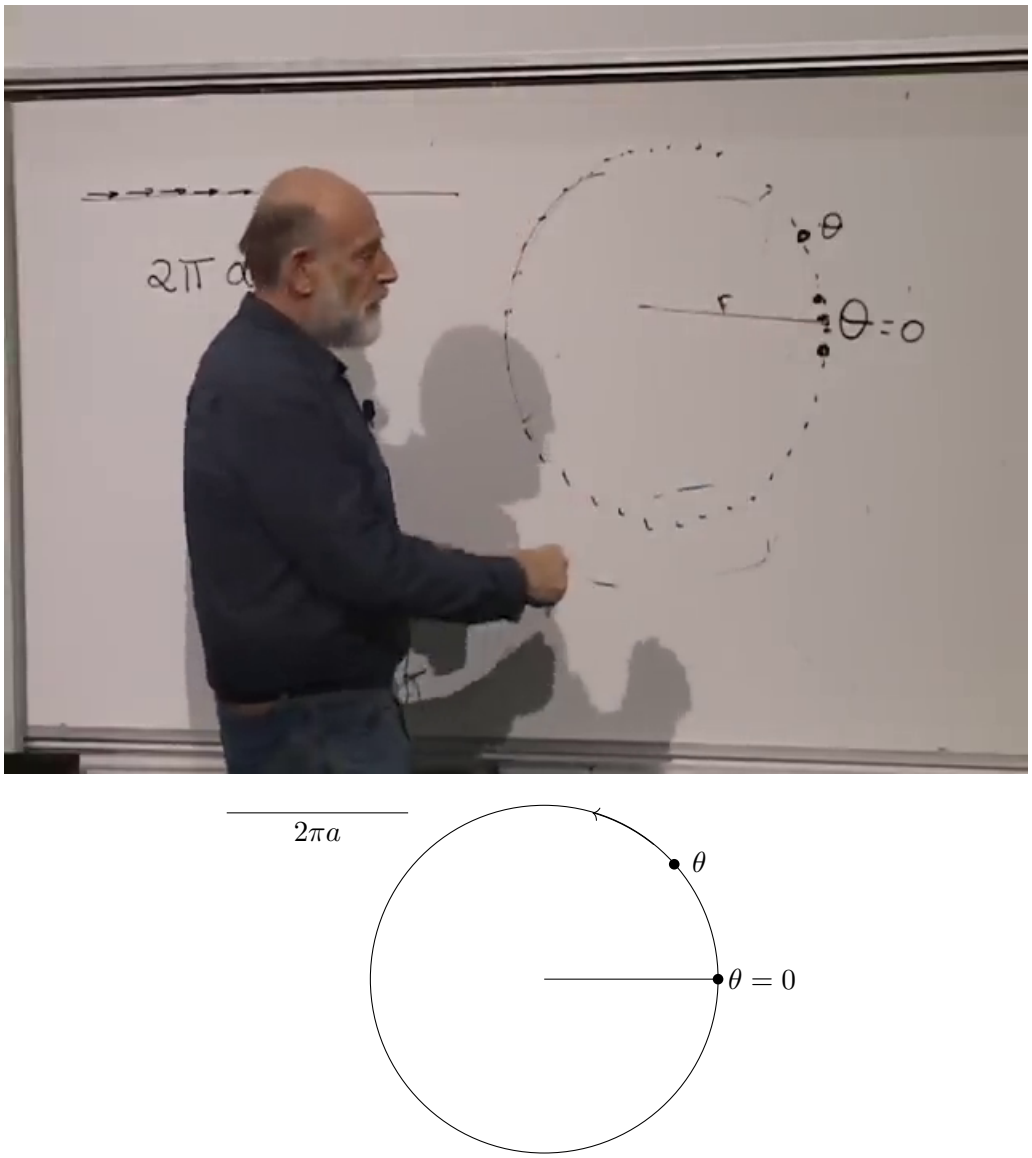


Figure 1.4: The circle model of a closed one-dimensional universe. The original frame is retained because the angular labels and the separate $2\pi a$ segment are part of the lecture's geometric reasoning, not mere ornament.

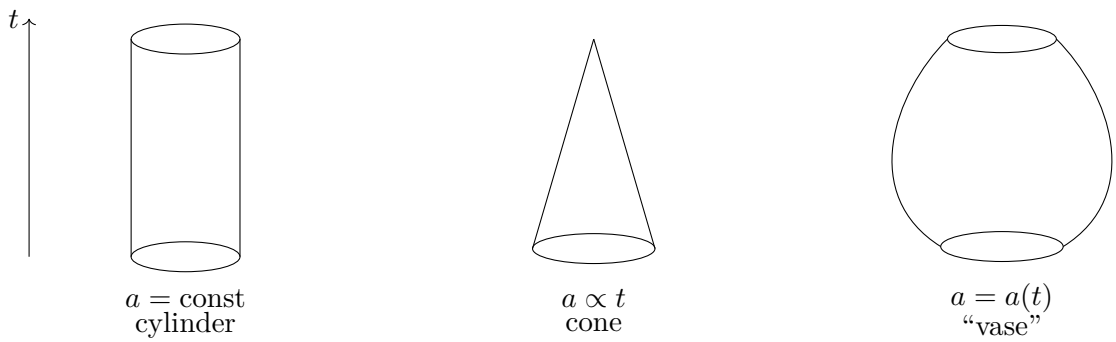


Figure 1.5: Explanatory redraw of the circle-universe spacetime for three choices of scale factor. These are external visualization aids that follow the lecture's narrative; they do not replace the intrinsic metric description.

This last equation is not directly visible in the captured frame, but it is the canonical cleaned form supported by the lecture's derivation.

Once the metric has been written, the analogy with the straight-line universe is immediate. If a depends on time,

$$d(t) = a(t)\,\Delta\theta, \tag{1.30}$$

and therefore

$$\dot{d} = \dot{a}\,\Delta\theta \tag{1.31}$$

$$= \frac{\dot{a}}{a}\,d. \tag{1.32}$$

So Hubble's law survives in the closed universe too, at least for separations small enough that we do not worry about the ambiguity of "the other way around" the circle.

The spacetime metric of the time-dependent circle model is then

$$d\tau^2 = dt^2 - a^2(t)\,d\theta^2. \tag{1.33}$$

The lecture visualizes this spacetime by stacking circles along a vertical time axis. If a is constant, the stack is a cylinder. If $a \propto t$, it is a cone. If a grows and shrinks in some more complicated way, the surface becomes vase-like.

1.7.1 Question & Answer

Question. Where did the Big Bang happen, and what do the cylinder and cone pictures really mean?

Answer. In this model the Big Bang is not an explosion from one privileged point in some ambient space. If $a(t) \to 0$, then every separation $a(t)\,\Delta\theta$ tends to zero. The whole closed universe shrinks everywhere at once. That is why Susskind answers the question "Where did it happen?" with "Everywhere."

The lecture makes this point concrete by imagining many galaxies spread around the circle. As we run time backward, the galaxies become more and more crowded. They are not converging toward

one special place on the circle. Rather, every distance between neighboring comoving points is shrinking simultaneously. The geometry itself is collapsing.

The second part of the answer concerns curvature. A cylinder drawn in three dimensions looks curved, but in the one-space, one-time toy model of the lecture it is intrinsically flat: one can cut it open and lay it out on a plane without stretching it. That is the case $a = \text{const}$. A cone corresponds to $a \propto t$. A cone is also flat except at its tip, where there is a conical singularity. So the tip marks a special spacetime event, but the ordinary part of the cone is flat in exactly the same intrinsic sense.

The lesson is that one must not read curvature naively from an embedding diagram. The cylinder, cone, and vase pictures are pedagogical devices for visualizing the time dependence of the circle universe. The intrinsic spacetime geometry is encoded by the metric, and the Big Bang, in this closed model, is an everywhere-event in that geometry.

1.8 Summary

The lecture unfolds a single idea through increasingly richer settings. We begin with comoving labels on a line and discover that physical distance is $d = a\,\Delta x$. Differentiating at fixed label separation gives Hubble's law, $\dot{d} = (\dot{a}/a)d$. That same scale factor then reappears in the spatial metric, $ds^2 = a^2 dx^2$, and finally in the spacetime metric, $d\tau^2 = dt^2 - a^2 dx^2$, or in three dimensions $d\tau^2 = dt^2 - a^2(dx^2 + dy^2 + dz^2)$.

Homogeneity and isotropy justify the extreme simplicity of the metric on large scales, though only as empirical and scale-dependent facts within the observable universe. The closing circle model then shows how a universe may be closed and bounded without boundary, while still obeying a local Hubble law. The payoff is conceptual as much as mathematical: cosmic expansion is not matter flying out from one point into pre-existing space. It is the geometry of space itself changing its scale everywhere at once.

Chapter 2

Cosmology Lecture 2: From the Expanding Metric to Friedmann's Equation

We were talking last time about the metric of space. This lecture picks up exactly there and asks the next question: what equation tells the scale factor how to change with time? Susskind announces the plan immediately. First a little geometry, then dynamics; and the dynamics, at least for now, will be derived not from general relativity but from Newton's equations. That choice gives the lecture its shape. We begin with an expanding comoving grid, pass through a deliberately simple projectile problem, and then watch that one-body Newtonian problem turn into Friedmann's equation. The selected board frames are kept below as local mathematical evidence, following Leonard Susskind's lecture with supporting curation from LazyingArt LLC and Video2Book.

2.1 From Homogeneity to the Expanding Metric

The standing assumptions are homogeneity and isotropy. Under those assumptions Hubble's law is not something tacked on from observation after the fact; it is the natural kinematics of a uniformly expanding universe. At a fixed time,

$$v = H(t)\, d. \tag{2.1}$$

The proportionality is linear in the distance, and the important qualification comes immediately: the so-called Hubble constant is constant in space, not in time. Homogeneity means it does not vary from place to place. It does *not* mean it remains unchanged as the universe evolves.

Before writing any metric, Susskind pauses over the question of measuring rods. If space expands, why do meter sticks not expand? The lecture's answer is concrete. The forces binding ordinary matter are enormously stronger than the tiny effect of cosmological expansion on atomic scales. One may imagine adding a minuscule extra repulsion between atoms; it would shift an equilibrium separation a little, but it would not tear apart a tightly bound rod. The cosmological expansion is therefore seen not in bound objects but in the growing separation of sufficiently distant, sufficiently unbound systems such as galaxies.

This is the reason for comoving coordinates. We assign coordinates that move with the Hubble flow. A galaxy with no extra peculiar motion then keeps the same coordinate value for all time. What

changes is the relation between coordinate separation and physical distance measured in meters. That changing conversion factor is the scale factor $a(t)$.

For flat space the spatial line element is

$$ds^2 = a(t)^2 \left(dx^2 + dy^2 + dz^2\right). \tag{2.2}$$

Here x, y, z are comoving coordinates. If two galaxies have a fixed coordinate separation Δx along one axis, then their physical distance is

$$d(t) = a(t)\,\Delta x. \tag{2.3}$$

Differentiating with respect to time gives

$$v = \dot{d} = \dot{a}\,\Delta x = \frac{\dot{a}}{a}\,d. \tag{2.4}$$

So once the comoving metric is in hand, the Hubble parameter becomes

$$H(t) = \frac{\dot{a}}{a}. \tag{2.5}$$

That is already the first bridge from geometry to dynamics.

2.2 Flat Spatial Slices Inside Curved Spacetime

Now we shift from the metric of space to the metric of spacetime. The lecture insists on an important point: three-dimensional space may be flat while spacetime is nevertheless curved. If the geometry of space changes with time, then spacetime is no longer Minkowski even if each spatial slice remains flat. Susskind's image is a flat rubber sheet being stretched by distant giants. The sheet stays flat as a sheet, but the four-dimensional history is curved.

Starting from the special-relativistic form $d\tau^2 = dt^2 - ds^2$, we upgrade the spatial metric to

$$d\tau^2 = dt^2 - a(t)^2 \left(dx^2 + dy^2 + dz^2\right). \tag{2.6}$$

The coefficients do not depend on position in space; they depend only on time. That is the metric expression of homogeneity. In matrix language, completing the spoken board description in the standard way, one writes

$$g_{\mu\nu} = \mathrm{diag}\Big(1, -a(t)^2, -a(t)^2, -a(t)^2\Big). \tag{2.7}$$

The off-diagonal terms vanish. There are no $dx\,dy$ or $dx\,dt$ terms. The metric is almost as simple as that of flat spacetime, except that the conversion between spatial separation and time is now time-dependent.

The picture is not decoration. It carries the lecture's intuition. Alice and Bob remain one coordinate unit apart, but one coordinate unit comes to represent more and more physical distance as time goes on. If $a(t)$ were driven to zero, then every physical separation would be driven to zero with it. In the lecture this is said in plain language: everything gets squashed on top of everything else, and the density becomes infinite. If $a(t)$ grows, the universe becomes correspondingly more diffuse.

There is another subtle point here. This metric has the stated form only in the frame moving with the galaxies. Rotations leave the spatial part unchanged, but a Lorentz boost does not preserve the same isotropic appearance. Matter itself picks out the preferred comoving frame.

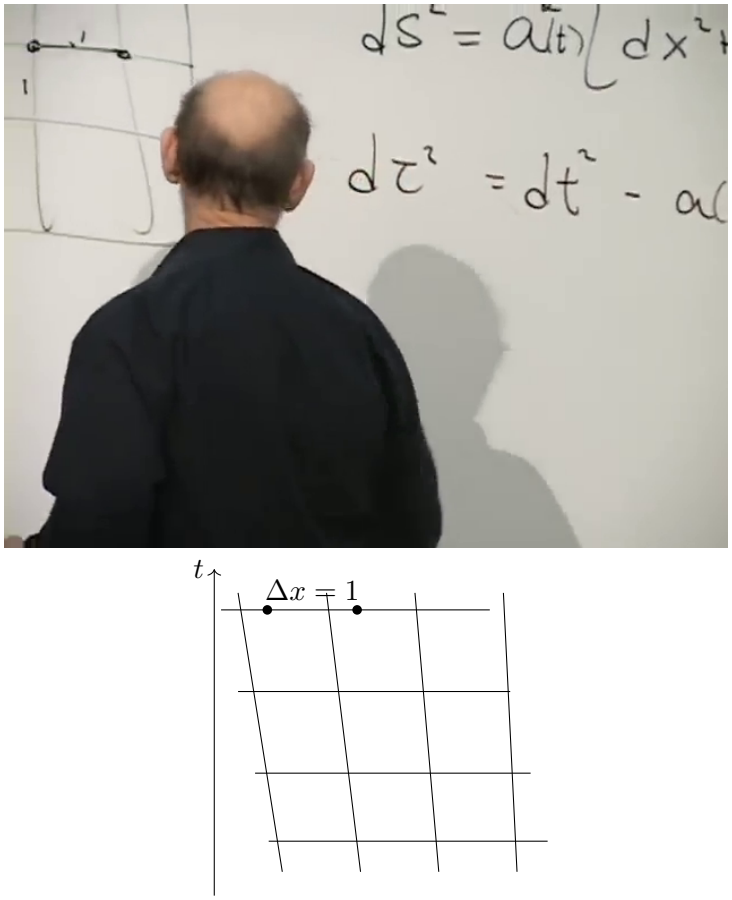


Figure 2.1: Spacetime interval and expanding-grid sketch. The lecture frame is retained as board evidence; the redraw beneath it makes the comoving grid legible as a geometric cartoon of fixed coordinate separation and increasing physical distance.

2.2.1 Question & Answer

Whose time is t in $a(t)$? It is the time measured by clocks moving with the galaxies. The lecture even says: imagine a cosmic clock factory making identical clocks, one carried by each galaxy. The coordinate time t is the time those comoving clocks measure.

This is also immediate from the metric. Along a galaxy worldline in the comoving frame we have

$$dx = dy = dz = 0, \tag{2.8}$$

so the interval reduces to

$$d\tau^2 = dt^2, \tag{2.9}$$

hence

$$dt = d\tau. \tag{2.10}$$

So the time appearing in $a(t)$ is cosmic proper time along the flow lines of the galaxies.

2.3 Newtonian Warm-Up: A Projectile in a Gravitational Field

With the geometry fixed, Susskind makes the lecture's second major pivot. Rather than deriving the dynamics of $a(t)$ from general relativity, he cools the problem down first and goes back to a familiar Newtonian one-body system. Before doing cosmology, let us understand a particle ejected radially outward from a heavy gravitating mass.

Take a heavy mass M and a much smaller mass m. Give the small mass an outward impulse and then let it move freely under gravity. Angular complications are not the point here, so the motion is taken to be radial. What matters for the cosmological analogy is energy conservation.

The kinetic energy is

$$\frac{1}{2}mv^2. \tag{2.11}$$

The gravitational potential energy has the product of the masses, Newton's constant, and one power of the distance in the denominator:

$$-\frac{GMm}{d}. \tag{2.12}$$

On the board the factors are written in the order mMG/D; in the cleaned equations we restore the conventional form GMm/d, but the board order is kept visible in the screenshots below. The conserved total energy is then

$$E = \frac{1}{2}mv^2 - \frac{GMm}{d}. \tag{2.13}$$

Here "constant" means constant in time for the particular particle under discussion. Different projectiles may of course have different values of E.

Notice how the lecture builds the formula. First the kinetic term is written. Then the potential term is added. Only after that does the negative sign become the issue. The sequence matters because later the cosmological argument will lean on the same intuition: not just on the existence of a formula, but on what the sign of the gravitational term means.

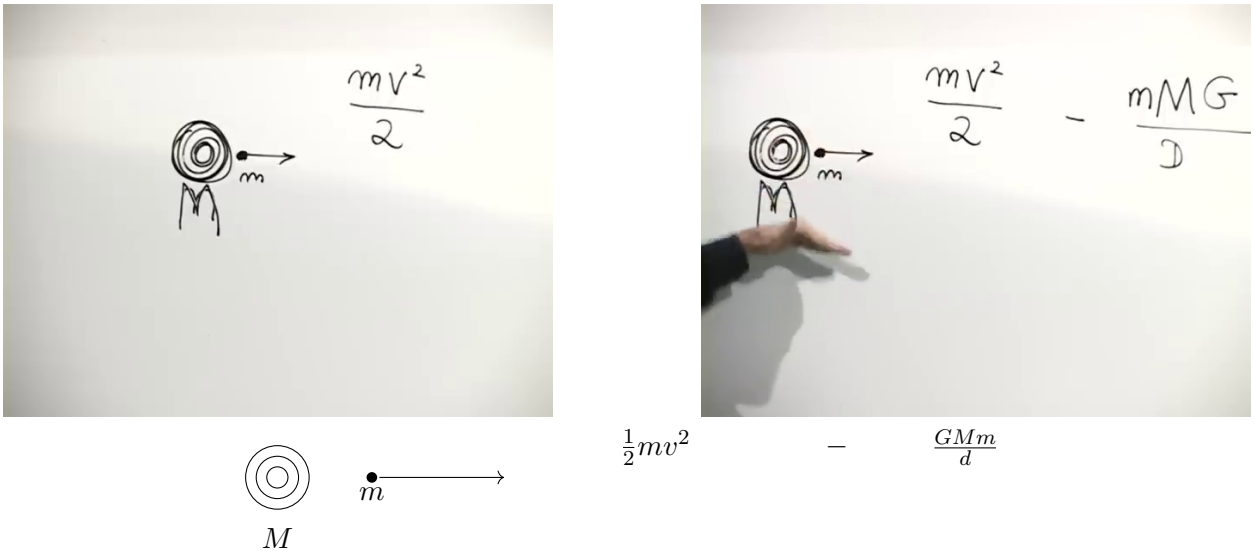


Figure 2.2: The Newtonian warm-up in the same order as the board. First the kinetic term, then the full energy balance with negative gravitational potential. The redraw regularizes the source–test-mass geometry without replacing the lecture frames.

2.3.1 Question & Answer

Why is the gravitational potential energy negative? The lecture does not answer this by abstract convention. It answers it by drawing the graphs of $1/D$ and $-1/D$. Forget the constants. The function $1/D$ decreases as D increases. The function $-1/D$ is the same curve upside down. Forces push in the direction of decreasing potential energy. Therefore an attractive force, which pulls inward toward smaller D, must correspond to the upside-down curve:

$$V(d) \propto -\frac{1}{d}. \tag{2.14}$$

Bringing the mass closer to the source creates a larger amount of negative potential energy. Moving it farther away makes the energy less negative. That is exactly what attraction means in energy language.

2.4 Escape Velocity and the Three Energy Regimes

Once the conserved energy is on the board, the lecture does not calculate right away. Instead it classifies the possible motions. There are only three cases.

- If $E < 0$, the motion is bound. The particle rises, slows, turns around, and falls back.
- If $E = 0$, the particle has exactly the escape velocity. The distance keeps increasing forever, while the speed decreases toward zero.
- If $E > 0$, the particle escapes completely. Gravity still decelerates it for a while, but not enough to prevent a nonzero asymptotic outward speed.

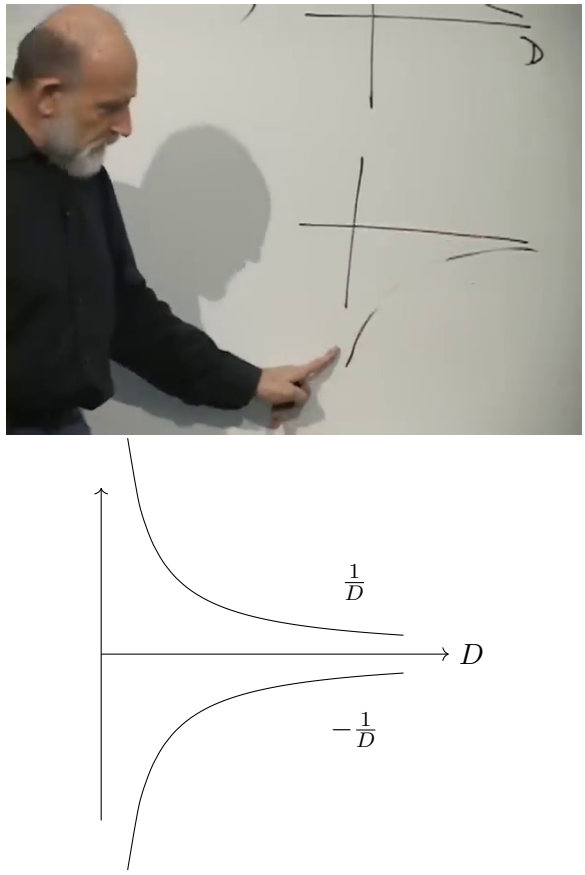


Figure 2.3: Qualitative potential-energy graphs. The lecture frame is kept because it records the actual board argument; the redraw regularizes the axes and the two comparison curves $1/D$ and $-1/D$.

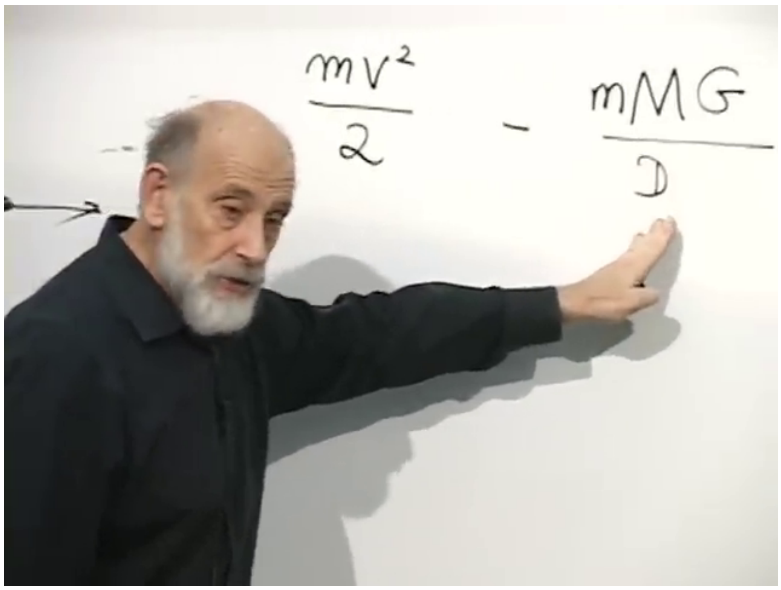


Figure 2.4: Later emphasis frame: at small enough initial outward speed, the negative potential term dominates the total energy.

The critical case $E = 0$ needs to be heard carefully. The distance is unbounded, but the speed goes to zero. It cannot actually come to rest at any finite time, because if it did, it would turn around and fall back. So the speed decreases forever without ever quite reaching zero at finite time.

For sufficiently small starting velocity the potential term dominates from the beginning, so the energy is negative:

This is the bridge to cosmology. The universe, in Newtonian language, behaves the same way. If the initial Hubble flow is large enough, galaxies escape from one another forever. If it is too small, gravity eventually wins and the universe recollapses. Between the two lies the critical case, the cosmological analog of escape velocity.

2.5 Newton's Shell Theorem and the Cosmological Reduction

The projectile analogy becomes a cosmological calculation only because of Newton's shell theorem. This is the lecture's third hinge, and Susskind takes time to make the audience trust it.

For a spherically symmetric mass distribution, the gravitational force on a particle at radius d is exactly the same as if all the mass inside radius d were concentrated at the center, while the mass outside radius d contributes no net force. The theorem is stronger than the familiar outside-a-sphere statement. A perfect spherical shell exerts no force on a particle anywhere inside it, even off-center. This is a surprising property, and in the lecture Susskind reminds us that it is special to the inverse-square law.

He even sketches a water-flow analogy: a point source produces a $1/r^2$-type flux in three dimensions, and a symmetric shell of such sources produces no net interior flow. The details are postponed, but the moral is clear. Exterior shells cancel.

Now we use that theorem in cosmology. Pick any galaxy and, for the sake of calculation, call it the

center. Because the universe is homogeneous and isotropic, this choice cannot matter. Pick a second galaxy at comoving coordinate x. Draw a sphere centered on the first galaxy and passing through the second. Then only the mass inside that sphere matters for the motion of the second galaxy.

Two further facts are essential.

First, the center is arbitrary. We could have centered the calculation on Alice, Bob, Fred, or any other galaxy, and the same equation must come out.

Second, the sphere expands with the Hubble flow. It therefore encloses the same galaxies for all time, and so its enclosed mass is constant in time even though its physical radius changes.

2.6 From Galaxy Energetics to Friedmann's Equation

We are now ready to write for a galaxy exactly the same kind of Newtonian energy equation we wrote for the projectile. After dividing by the test mass, it is convenient to write

$$\frac{v^2}{2} - \frac{GM(d)}{d} = \varepsilon(x), \tag{2.15}$$

where $\varepsilon(x)$ is constant in time for the galaxy at comoving coordinate x. It is not required to be the same for every x.

The physical distance from the chosen center to that galaxy is

$$d = a(t)x, \tag{2.16}$$

and because x is comoving,

$$v = \dot{d} = \dot{a}\,x. \tag{2.17}$$

If $\rho(t)$ is the homogeneous density, then the enclosed mass is

$$M(d) = \frac{4\pi}{3}\rho\, d^3 = \frac{4\pi}{3}\rho\, a^3 x^3. \tag{2.18}$$

Substituting these into the energy equation gives

$$\frac{(\dot{a}x)^2}{2} - \frac{G\left(\frac{4\pi}{3}\rho\, a^3 x^3\right)}{ax} = \varepsilon(x), \tag{2.19}$$

$$\frac{\dot{a}^2 x^2}{2} - \frac{4\pi G}{3}\rho\, a^2 x^2 = \varepsilon(x). \tag{2.20}$$

Here the lecture lingers on the factor of x^2, and rightly so. Both terms on the left-hand side are proportional to x^2. Everything else there is the same for every galaxy. Therefore the right-hand side must also be proportional to x^2. That is the precise meaning of the seemingly casual statement that the "constant" must scale with x^2. The farther galaxies carry larger energies, and that is exactly what one should expect since they are also moving faster.

So we write

$$\varepsilon(x) = -\frac{k}{2}\,x^2, \tag{2.21}$$

where the normalization of k is conventional. Dividing through by $x^2/2$, we obtain

$$\dot{a}^2 = \frac{8\pi G}{3}\rho\, a^2 - k. \tag{2.22}$$

Dividing once more by a^2, we arrive at the standard historical form,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \tag{2.23}$$

This is Friedmann's equation in the Newtonian form emphasized in the lecture. One can also write it as

$$H(t)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}, \tag{2.24}$$

since $H = \dot{a}/a$. Susskind likes to pause here and point out the historical drama of the result: a purely Newtonian energy argument has landed on the same equation that later appears as a component of Einstein's equations.

The sign of k tracks the same three cases we already met in the escape-velocity problem. The critical case is $k = 0$. In units with $c = 1$, k is dimensionless.

2.7 Matter Density, the Critical Case, and the Age of the Universe

If we sent ourselves home at this point and tried to solve Friedmann's equation, the first objection would be immediate: ρ is not a constant. It changes as the universe expands. So the next task is to express that dependence cleanly.

Susskind does it with a unit comoving cube. Take a cube that is one unit on a side in coordinate space. The galaxies move with the comoving coordinates, so the same galaxies remain inside that cube for all time. Let the mass inside it be $m_\square$. Then the physical side length of the cube is $a(t)$, and the physical volume is $a(t)^3$. Therefore

$$\rho(t) = \frac{m_\square}{a(t)^3}. \tag{2.25}$$

Substituting this into Friedmann's equation gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\, m_\square}{3a^3} - \frac{k}{a^2}. \tag{2.26}$$

Now choose the simplest case, $k = 0$, the analog of exact escape velocity. Then

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\, m_\square}{3a^3}. \tag{2.27}$$

Before solving, the lecture points out a qualitative consequence. The right-hand side is always positive. Therefore the left-hand side cannot pass through zero, which means the model does not turn around. In the critical case the universe expands forever, but with a continually decreasing expansion rate.

Now comes the worked solution. Rather than grinding through the differential equation abstractly, Susskind tries a power law:

$$a(t) = ct^p. \tag{2.28}$$

Then

$$\dot{a} = cp\, t^{p-1}, \qquad \frac{\dot{a}}{a} = \frac{p}{t}. \tag{2.29}$$

Substituting into the equation gives

$$\frac{p^2}{t^2} = \frac{8\pi G\, m_\square}{3c^3\, t^{3p}}. \tag{2.30}$$

For this to hold for all t, the powers of t must match:

$$3p = 2. \tag{2.31}$$

So

$$p = \frac{2}{3}. \tag{2.32}$$

Then matching coefficients gives

$$\frac{4}{9} = \frac{8\pi G\, m_\square}{3c^3}, \tag{2.33}$$

hence

$$c^3 = 6\pi G\, m_\square. \tag{2.34}$$

The critical matter-dominated solution is therefore

$$a(t) = (6\pi G\, m_\square)^{1/3}\, t^{2/3}. \tag{2.35}$$

More generally one may absorb the remaining integration constant into a shift of the origin of time and write

$$a(t) \propto (t - t_0)^{2/3}. \tag{2.36}$$

This is a more interesting result than it may look at first sight. The universe does not expand linearly with time. Linear growth would be the analog of unaccelerated motion. Here the growth is slower than linear:

$$a(t) \propto t^{2/3}. \tag{2.37}$$

So the expansion slows down, but in the critical case it never turns around.

The interpretation of $t = 0$ is also part of the lecture's rhythm. If $a(t)$ really went to zero, everything would be squashed on top of everything else and the density would diverge. Susskind immediately softens the literalness of that statement: the Newtonian model cannot be trusted all the way down to infinite density. But it does tell us that the universe emerged from an extremely dense early state. That is the Big Bang limit of the model.

The Hubble parameter now becomes explicit:

$$H(t) = \frac{\dot{a}}{a} = \frac{2}{3t}. \tag{2.38}$$

So in this model

$$t = \frac{2}{3H}. \tag{2.39}$$

This is one of the lecture's payoffs. The Hubble constant is not a constant at all; it falls like $1/t$. Susskind jokes that a final examination could consist of a single question: *Is the Hubble constant a constant?* The correct answer is no.

The same formula turns a present-day measurement of H into an age estimate. In the lecture Susskind quotes an age of roughly 13 to 14 billion years and remarks that $k = 0$ is observationally very realistic, with no known significant deviation from flatness at the level of precision he quotes.

2.7.1 Question & Answer

Why does the age estimate not depend on the arbitrary fiducial box mass? The lecture's late discussion of this point is subtle because the notation becomes slippery, but the underlying idea is clean. The fiducial comoving box is conventional. If we double its side length, then the mass inside it grows by a factor of 8, while $m_\square^{1/3}$ grows by a factor of 2. The corresponding normalization of a changes by the same factor. What is physical is the density

$$\rho = \frac{m_\square}{a^3}, \tag{2.40}$$

or equivalently the invariant combination $a/m_\square^{1/3}$, not a and $m_\square$ taken separately.

The cancellation appears directly in the solution. If

$$a(t) = ct^{2/3}, \tag{2.41}$$

then

$$\dot{a} = \frac{2}{3} c\, t^{-1/3}, \tag{2.42}$$

so

$$\frac{\dot{a}}{a} = \frac{\frac{2}{3} c\, t^{-1/3}}{ct^{2/3}} = \frac{2}{3t}. \tag{2.43}$$

The constant c, and therefore the dependence on the fiducial-box mass, drops out of $H = \dot{a}/a$. That is why the age relation

$$t = \frac{2}{3H} \tag{2.44}$$

does not depend on the arbitrary normalization of the comoving box in the critical model.

This is also the right place to hear the lecture's last reinterpretation. Rearranging Friedmann's equation gives

$$\frac{\dot{a}^2}{a^2} - \frac{k}{a^2} = \frac{8\pi G}{3}\rho. \tag{2.45}$$

The left-hand side is geometric: it is built from the scale factor and, next time, from spatial curvature. The right-hand side is matter density, or in relativistic language energy density. That is exactly the sort of equation Einstein's theory ought to produce.

2.8 Summary

The lecture unfolds in a very deliberate sequence. Homogeneity and isotropy fix the kinematics of expansion. The comoving metric gives the scale factor a precise meaning. A Newtonian projectile problem provides the right intuition for conserved energy and escape velocity. Newton's shell theorem then converts the many-body universe into an effective one-body problem, and that reduction yields

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \tag{2.46}$$

In the critical matter-dominated case $k = 0$, the solution is

$$a(t) \propto t^{2/3}, \qquad H(t) = \frac{2}{3t}. \tag{2.47}$$

So the Newtonian model already gives a serious cosmological prediction: a slowing but never reversing critical expansion, together with an age estimate from the present Hubble parameter. The next lecture will reopen the equation from the relativistic side and show how curvature and Einstein's equations complete the story.